

TIAMAT Scene-Graph Exploration

^{*}Transfer from Imprecise and Abstract Models to Autonomous Technologies DARPA Challenge

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Abstract—Autonomous robots operating in novel environments must build semantic understanding of their surroundings while simultaneously deciding where to explore. Existing 3D scene graph methods assume indoor environments with natural hierarchies and require offline construction after full exploration, limiting their applicability to unstructured outdoor settings. We present a framework for online dynamic scene graph construction that actively guides frontier-based exploration. Our approach incrementally builds scene graphs from streaming RGBD data, dynamically updating object nodes and spatial relations as new observations arrive. A task parser extracts relevant object categories from natural language task descriptions, and an LLM-based frontier selector reasons over the partial scene graph to prioritize exploration targets. We demonstrate that integrating scene graph construction into the exploration loop enables robots to make semantically informed navigation decisions in open-ended environments without requiring prior maps or complete spatial coverage.

I. INTRODUCTION

Autonomous robots deployed in real-world settings must navigate and explore environments they have never seen before. Search and rescue operations require robots to traverse disaster sites while identifying survivors and hazards. Service robots must operate in cluttered household or workplace environments filled with diverse objects and changing layouts. Military reconnaissance tasks robots with surveying forward operating bases or unfamiliar terrain. In each case, the robot cannot rely on pre-built maps or prior exploration; it must build semantic understanding of its surroundings while simultaneously deciding where to go next.

This problem requires solutions to three interconnected challenges. First, robots need structured representations that capture the semantic richness of real environments, including objects, regions, and their relationships, and these representations must be constructed online as the robot moves. Second, robots must navigate and map unknown spaces that may be outdoor or open-ended, selecting where to explore next based on partial observations that could be incomplete or incorrect. Third, robots need to reason over long horizons, planning sequences of actions toward high-level goals using semantic understanding.

In this paper, we propose a framework for online dynamic scene graph construction and utilization that addresses these

challenges within the exploration loop of novel, unstructured environments.

II. RELATED WORKS

A. Open-Vocabulary Scene Graphs

Recent advances in 3D scene graph representations have enabled robots to capture rich semantic and spatial relationships for downstream tasks. Rosinol et al. introduced 3D Dynamic Scene Graphs (DSGs), providing a hierarchical representation encoding objects, places, rooms, and agents with spatio-temporal relations for planning and human-robot interaction [4]. ConceptGraphs extended this approach by using 2D vision-language foundation models to construct open-vocabulary 3D scene graphs without task-specific training, enabling generalization to novel semantic classes [1]. These representations offer compact, semantically rich alternatives to dense per-point feature maps.

However, existing hierarchical scene graph architectures are mainly designed for indoor environments with natural spatial hierarchies: floors containing rooms containing objects [4]. This structural assumption does not transfer to outdoor or unstructured environments lacking clear room-level organization. Furthermore, scene graph construction in these works typically occurs offline after full environment exploration, requiring complete spatial coverage before the representation becomes useful for planning.

B. Online Frontier Exploration

Classical frontier-based exploration methods systematically expand known regions by navigating to boundaries between explored and unexplored space. These approaches typically select frontiers based on geometric criteria such as information gain, travel cost, or expected map uncertainty reduction. While effective for occupancy map construction, such methods operate independently of semantic scene understanding and do not use structured representations to prioritize exploration targets based on task-relevant context.

Recent work has begun adding semantic information into exploration, but scene graphs remain underused in guiding where to explore. Systems like DovSG [2] and Hi-Dyna Graph [3] update scene graphs during operation, yet these updates serve to maintain an accurate world model rather

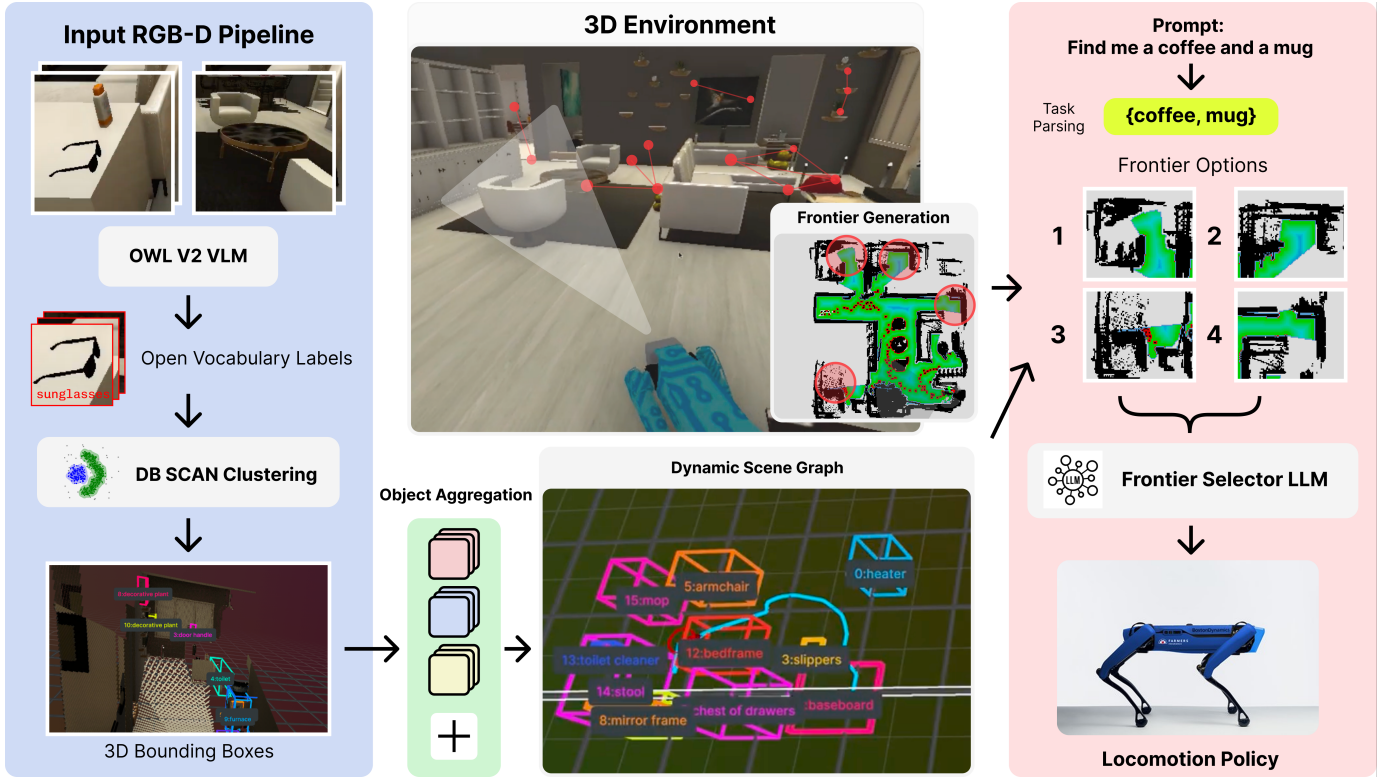


Fig. 1. Enter Caption

than to actively inform frontier selection. The graph acts as a passive knowledge base built alongside exploration rather than a representation that affects the exploration strategy itself.

C. LLM Navigation and Planning with Scene Graphs

Large language models have shown strong reasoning capabilities for robotic task planning when grounded in structured scene representations. MoMa-LLM grounds LLM reasoning in open-vocabulary scene graphs for zero-shot mobile manipulation [5]. DovSG combines scene graphs with language-guided planning for long-term task execution in changing environments [2]. Hi-Dyna Graph uses LLM agents to interpret hierarchical graphs, infer task triggers, and generate executable robotic instructions [3].

These approaches show that scene graphs provide effective interfaces between language models and robotic action spaces. However, they assume scene graphs constructed from thorough exploration, with mechanisms for correction through repeated observation. The planning modules operate over largely complete representations rather than actively reasoning about where to explore next based on partial knowledge.

D. Contributions

We present a framework for online dynamic scene graph construction and use for frontier-based exploration in outdoor environments. Our approach consists of four main components:

- 1) An efficient algorithm for online scene graph generation from RGBD data that dynamically updates, adds,

merges, and removes object nodes as new views are captured. Relational edges are created through spatial proximity and checked via parallelized LLM calls, and we use anchor points rather than strict hierarchical structures to handle unstructured outdoor scenes.

- 2) A task parser that extracts relevant object categories and semantic keys from natural language task descriptions.
- 3) An LLM-based frontier selector that reasons over the partial, incrementally built scene graph to choose exploration targets.
- 4) A locomotion policy that navigates the robot to the selected frontier.

Together, these components enable robots to actively use incomplete scene graph representations to guide exploration in open-ended environments.

III. METHOD

A. Online Scene Graph Generation

We construct and maintain a scene graph incrementally from streaming RGBD data captured by five onboard cameras. At each timestep, we run OWLv2 [6], an open-vocabulary vision-language model, on each camera image with a minimum confidence threshold τ_c . For each detection, we project the corresponding depth pixels into a 3D point cloud using known camera intrinsics and robot pose. Since background clutter can skew bounding box and centroid calculations, we apply DBSCAN clustering [7] to isolate object points. The resulting

point cloud is voxelized for efficient storage and comparison, from which we compute the 3D bounding box and centroid.

To track objects across frames, we maintain a set of tracked objects and compare each new detection against existing objects within a centroid proximity threshold. For each candidate match, we compute a similarity score:

$$s = w_g \cdot \text{sim}_{\text{geo}}(c, b) + w_v \cdot \text{sim}_{\text{voxel}}(V, V') + w_e \cdot \text{sim}_{\text{emb}}(e, e') \quad (1)$$

where sim_{geo} compares centroids c and bounding boxes b , $\text{sim}_{\text{voxel}}$ measures 3D voxel overlap, and sim_{emb} computes cosine similarity between VLM embeddings. If s exceeds threshold τ_s , we append the detection to the existing object; otherwise, we create a new object.

An object becomes confirmed after accumulating n or more detections. Rather than permanently assigning a label from the first observation, we determine the label through a weighted vote over the k most recent detections:

$$\ell^* = \arg \max_{\ell} \sum_{i=1}^k c_i \cdot \mathbb{I}[\ell_i = \ell] \quad (2)$$

where c_i and ℓ_i are the confidence and label of detection i . This allows labels to update as the robot obtains better views. To handle hallucinated detections from distant or partial views, we remove objects that remain within the camera field of view and unoccluded yet fail to match any detection for m consecutive frames.

When a new object is confirmed, we query a 2D spatial index to find nearby objects within distance d and compute initial spatial relations (above, below, next to, on, in) based on centroid geometry. These edges are verified asynchronously by an LLM that checks semantic plausibility given the object labels.

B. Task Parser

The task parser converts high-level task descriptions into object queries for the scene graph. Given a natural language task such as “find shelter from the rain,” the parser extracts relevant object categories (e.g., tent, tarp, building) and generates synonyms for each to expand the open-vocabulary detector’s search space.

C. LLM Frontier Selector

The frontier selector identifies candidate exploration targets and uses the partial scene graph to choose among them. We first extract frontiers by identifying unexplored navigable regions from the occupancy map, applying morphological operations to reduce noise, and detecting boundaries where explored space meets unexplored space using Canny edge detection. Frontier pixels are grouped into clusters via connected components, filtered by minimum size, and merged if within 0.2m to reduce fragmentation.

For each candidate frontier, we query the scene graph for nearby confirmed objects and their edges to build a local context description. We then prompt an LLM with the frontier descriptions, surrounding scene graph context, and target

objects from the task parser. The LLM selects the frontier most likely to yield task-relevant discoveries given the partial semantic information available.

D. Locomotion Policy

Once a frontier is selected, a heuristic locomotion policy navigates the robot to the target location, taking the current robot state and goal position as input and outputting velocity commands.

IV. IMPACT AND FUTURE WORK

Our framework opens several directions for future research. First, because scene graphs provide a structured, compact representation, they are well suited for real-time sharing between multiple robots. A team of robots could jointly build and query a shared scene graph, enabling coordinated exploration of large environments. Second, extending the scene graph to incorporate temporal reasoning would allow robots to track how environments change over time, such as objects being moved or doors opening, which is important for long-duration deployments. Third, the structured nature of the representation makes it amenable to human-in-the-loop corrections, where an operator could add, remove, or relabel objects to improve the scene graph during operation.

V. LIMITATIONS

Our approach has several limitations. LLM calls for edge verification and frontier selection introduce latency and computational cost, which may not scale to settings requiring very fast exploration. While open-vocabulary detection enables generalization to novel objects, it still struggles with uncommon objects and can confuse visually similar categories. Finally, our current system lacks a mechanism for revisiting uncertain objects to confirm or correct initial observations, meaning early errors can persist in the scene graph.

VI. CONCLUSION

We addressed the problem of autonomous exploration in novel outdoor environments, where robots must build semantic understanding while simultaneously deciding where to explore. Existing scene graph methods assume indoor hierarchies and offline construction, leaving a gap for unstructured, open-ended settings. We presented a framework that constructs dynamic scene graphs online from RGBD data, parses high-level tasks into object queries, and uses an LLM to select frontiers based on partial semantic information. Our approach enables robots to actively use incomplete scene graph representations to guide exploration rather than treating the graph as a passive byproduct. As scene graph methods continue to mature, we believe tighter integration with exploration and planning will be essential for deploying robots in complex real-world environments.

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APPENDIX

A. Edge Verification Prompt

Given two objects in a 3D scene:

Object A: source_desc

Object B: target_desc

Geometric context: Object A is geometric_context
Object B.

What is the most likely spatial relationship? Choose ONE from:

- on (A is on top of B)
- inside (A is inside B)
- next_to (A is beside B)
- above (A is above B but not on)
- below (A is below B)
- near (A is close to B)

Reply with ONLY the relationship word, nothing else.